

On Calibration of Modern Neural Networks

Chuan Guo^{*1} Geoff Pleiss^{*1} Yu Sun^{*1} Kilian Q. Weinberger¹

Abstract

Confidence calibration – the problem of predicting probability estimates representative of the true correctness likelihood – is important for classification models in many applications. We discover that modern neural networks, unlike those from a decade ago, are poorly calibrated. Through extensive experiments, we observe that depth, width, weight decay, and Batch Normalization are important factors influencing calibration. We evaluate the performance of various post-processing calibration methods on state-of-the-art architectures with image and document classification datasets. Our analysis and experiments not only offer insights into neural network learning, but also provide a simple and straightforward recipe for practical settings: on most datasets, *temperature scaling* – a single-parameter variant of Platt Scaling – is surprisingly effective at calibrating predictions.

1. Introduction

Recent advances in deep learning have dramatically improved neural network accuracy (Simonyan & Zisserman, 2015; Srivastava et al., 2015; He et al., 2016; Huang et al., 2016; 2017). As a result, neural networks are now entrusted with making complex decisions in applications, such as object detection (Girshick, 2015), speech recognition (Hannun et al., 2014), and medical diagnosis (Caruana et al., 2015). In these settings, neural networks are an essential component of larger decision making pipelines.

In real-world decision making systems, classification networks must not only be accurate, but also should indicate when they are likely to be incorrect. As an example, consider a self-driving car that uses a neural network to detect pedestrians and other obstructions (Bojarski et al., 2016).

^{*}Equal contribution, alphabetical order. ¹Cornell University. Correspondence to: Chuan Guo <cg563@cornell.edu>, Geoff Pleiss <geoff@cs.cornell.edu>, Yu Sun <ys646@cornell.edu>.

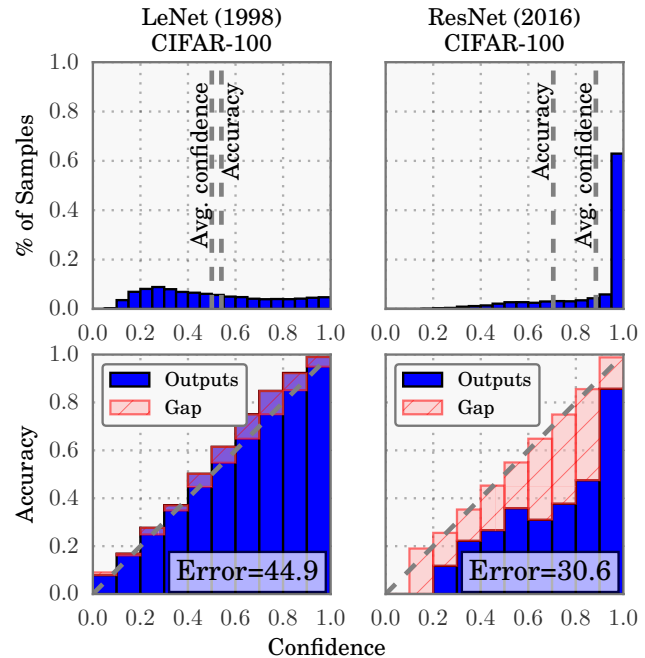


Figure 1. Confidence histograms (top) and reliability diagrams (bottom) for a 5-layer LeNet (left) and a 110-layer ResNet (right) on CIFAR-100. Refer to the text below for detailed illustration.

If the detection network is not able to confidently predict the presence or absence of immediate obstructions, the car should rely more on the output of other sensors for braking. Alternatively, in automated health care, control should be passed on to human doctors when the confidence of a disease diagnosis network is low (Jiang et al., 2012). Specifically, a network should provide a *calibrated confidence* measure in addition to its prediction. In other words, the probability associated with the predicted class label should reflect its ground truth correctness likelihood.

Calibrated confidence estimates are also important for model interpretability. Humans have a natural cognitive intuition for probabilities (Cosmides & Tooby, 1996). Good confidence estimates provide a valuable extra bit of information to establish trustworthiness with the user – especially for neural networks, whose classification decisions are often difficult to interpret. Further, good probability estimates can be used to incorporate neural networks into other probabilistic models. For example, one can improve performance by combining network outputs with a lan-